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CO1 - AT1

1) Data science can analyze student's attendance, previous marks, assignment scores, and learning activity to identify students who are at risk of failing. predictive models can be used to provide early warnings and targeted support.

2) Data Science can analyze any application of recommendation Systems. Data Science analyzes customers' previous purchases, Searches, ratings, and preferences to recommend products that are likely to interest them.

3) Data Science can analyze customer purchase history, complaints, usage patterns, and interactions to identify customers who may leave the company. This is called customer churn prediction, which helps the company take preventive actions.

4) Data Science can analyze historical sales data from previous festivals to identify products with high demand. The Supermarket can use these insights to plan inventory and stock popular products appropriately

5) Data Science can analyze soil conditions, weather data, rainfall, temperature, and crop history to recommend suitable crops and farming practices. This can help the farmer improve crop yield.

6) The facet involved is velocity. Velocity refers to the speed at which data is generated, collected, and processed. Here, temperature data is continuously updated every minute

7) Customer comments on social media are generally unstructured data because they do not follow a fixed tabular format. They may contain text, emojis, images, and other forms of information.

8) The data has data quality issues, specifically missing values and inaccurate/incorrect data. These problems must be cleaned and corrected before analysis to ensure reliable results.

9) The component used is HDFS (Hadoop Distributed File System). HDFS is a distributed storage system used to story very large datasets across multiple computers.

10) Apache kafka is suitable for collecting and handling live sensor data. It is a distributed streaming platform that can receive and process large amounts of real-time data from IoT devices.